Executive Summary

WW Elevators is requesting an upgrade to its current elevator system so the cab can support sideways movement in addition to normal vertical travel. Since that changes the way users interact with the system, the elevator control panel also needs to be redesigned. This project focuses on creating a new panel that allows users to control both vertical and horizontal movement while still keeping the system simple, safe, and easy to understand.

The concept for this design is based on U.S. Patent No. 10,696,521, which describes an elevator system that can move across vertical and horizontal guide rails through the use of rotating rail segments. That idea makes sideways travel possible, though the panel design has to make that movement practical for everyday users. The main purpose of this document is to define the system requirements, performance expectations, design limits, and project scope for the proposed panel.

System Requirements

The sideways elevator panel must support both standard elevator functions and the new sideways travel feature. Users should still be able to select floors the same way they would on a normal elevator, though the system must now also provide controls for horizontal movement. The panel should clearly communicate the elevator's current location and travel direction so users are not confused while using it.

The system must include these core functions:

- Floor selection buttons for vertical travel
- Left and right directional buttons for sideways travel
- A display showing current elevator position
- Door open and door close buttons
- An alarm button
- An emergency stop button

The system must also verify all user input before allowing the elevator to move. Since this panel controls more than one type of movement, the software has to reject unsafe or conflicting commands. For example, opposite directional inputs should not be accepted at the same time. The panel should also provide visible or audible feedback when a button is pressed and when the elevator reaches the selected destination. That kind of feedback matters because it helps the user know the system is working as expected. **Performance Requirements**

The panel should respond fast enough that users do not feel a delay between pressing a button and seeing a result. A slow response would make the whole system feel unreliable, so user input should be processed within one second or less. The panel also needs to perform consistently during repeated daily use without system failure, missed inputs, or noticeable lag.

The main performance expectations are:

- Input response time of one second or less
- Reliable operation during repeated daily use
- Immediate response from alarm and emergency controls
- Real time updates on the position display during movement

Safety features must always take priority. The emergency stop and alarm controls should respond immediately every time they are used. The display should also update continuously while the elevator is moving so passengers can track both floor position and sideways direction or section. Since this is not a normal elevator setup, real time status updates help reduce confusion and make the system easier to trust. **Design Constraints**

The sideways elevator panel must fit within the physical space of a standard elevator control panel. That means the design cannot be overloaded with buttons or visual clutter. The added controls have to fit into the layout in a way that still feels organized and readable. This project is also an upgrade to an existing system, so the panel should work with current elevator hardware and safety systems as much as possible rather than requiring a complete redesign.

The main design constraints include:

- Limited space on the control panel
- Compatibility with existing hardware
- Compatibility with current safety systems
- Clear button labels and visible indicators
- Accessibility for a wide range of users
- Reasonable cost for an upgraded system

Usability is one of the biggest concerns in this design. Sideways movement adds a new layer of complexity, so the panel has to stay intuitive. Directional controls should be clearly separated from normal floor controls, and emergency buttons should remain easy to locate and access at all times. The design should improve system capability without making users stop and figure out what every button means before they can use the elevator. **Schedule and Budget**

The estimated timeline for this project is about five to six weeks. The first week would focus on project planning and requirements analysis. After that, one to two weeks would be used to design the panel layout and create supporting CASE diagrams.
Development of the control logic and panel behavior would take about two weeks. The final week would be used for testing, revisions, and final review. **Estimated Timeline**

- Planning and requirements analysis: 1 week
- Design and diagrams: 1 to 2 weeks
- Development: 2 weeks
- Testing and revisions: 1 week

Total Estimated Duration: 5 to 6 weeks

The estimated budget for the project is approximately $12,000. This amount includes hardware updates, software and control development, and testing. Since this is a classroom project, the budget should be viewed as a realistic estimate rather than a final business quote. In a real world setting, the actual cost could change depending on hardware compatibility, safety certification needs, and the amount of engineering required to support sideways travel.
Estimated Budget

- Hardware updates: $4,000
- Software development: $6,000
- Testing: $2,000

Total Estimated Budget: $12,000

Appendices:

Located in the Appendix Directory:

Appendix/Appendix_A_Current_Use_Case.pdf

Appendix/Appendix_B_Propsed_Use_Case.pdf

Appendix/Appendix_C_Propsed_Flow_Chart.pdf

Current System

The current elevator panel supports only traditional vertical movement. Users can select floors, open and close the doors, activate the alarm, and use the emergency stop when needed.

The system is designed entirely around upward and downward travel, so there are no controls for sideways movement.

Proposed System

The proposed system upgrades the current panel by adding controls for horizontal travel. In addition to normal floor selection, the panel will include left and right directional buttons. The display will show both floor location and sideways position when needed. The software will verify user input before movement begins and reject any unsafe or conflicting commands. This helps the system stay safe while handling more complex movement options.

Suggested Diagrams

The appendices should include diagrams that show both the current system and the proposed system. These visuals would make the project easier to understand and would help show how the upgraded panel changes the flow of user interaction.

Suggested diagrams include:

- A use case diagram
- A simple system diagram

References

Inventor Unknown. (2020, June 30). *Elevator system* (U.S. Patent No. 10,696,521). United States Patent and Trademark Office.